

# Statistics - Set 1 - Solutions

Specification M6.1-M6.4.

## Answer key

| Question | Answer | Topic      |
|----------|--------|------------|
| 1        | A      | Statistics |
| 2        | C      | Statistics |
| 3        | C      | Statistics |
| 4        | F      | Statistics |
| 5        | F      | Statistics |
| 6        | A      | Statistics |
| 7        | D      | Statistics |
| 8        | F      | Statistics |
| 9        | D      | Statistics |
| 10       | D      | Statistics |
| 11       | H      | Statistics |

## Worked solutions

### Question 1 STA-01 · Statistics · spec M6.3

The mean of five numbers is 12. When a sixth number is added the mean becomes 14. What is the sixth number?

**Answer: A**

Total goes from 60 to 84: the sixth number is 24.

#### Full method - how and why

Means questions are almost always really about **totals**. Convert each mean into a total before doing anything else.

- 1 Original total  $= 5 \times 12 = 60$ .  
why:  $\text{mean} = \frac{\text{total}}{\text{how many}}$ , so  $\text{total} = \text{mean} \times \text{how many}$ .
- 2 New total  $= 6 \times 14 = 84$ .  
why: There are now six numbers, and their mean is 14.
- 3 The sixth number is the difference:  $84 - 60 = 24$ .  
why: Nothing else changed, so the whole increase in the total is the new number.

#### Where it goes wrong

- Answering 14 - the new **mean**, not the new **number**.
- Answering 84, the new total rather than the added value.
- Thinking the new number must be 2 more than the old mean; the mean rose by 2 for **all six**, so the newcomer must carry  $6 \times 2 = 12$  extra above the old mean of 12, giving 24.

**Takeaway** Turn means into totals; the added value is the change in total.

### Question 2 STA-02 · Statistics · spec M6.3

A histogram has a bar for the class  $10 \leq x < 15$  of height (frequency density) 3.2 and a bar for  $15 \leq x < 25$  of height 1.5. What is the total frequency of these two classes?

**Answer: C**

Frequency = density  $\times$  width:  $3.2 \times 5 + 1.5 \times 10 = 16 + 15 = 31$ .

#### Full method - how and why

On a histogram the **area** of a bar is the frequency, not its height. Height is frequency density.

- 1 Read the first bar: width  $15 - 10 = 5$ , height 3.2.  
why: Class widths on a histogram are usually unequal - that is why density is used.
- 2 Frequency = density  $\times$  width  $= 3.2 \times 5 = 16$ .  
why: Rearranging density  $= \frac{\text{frequency}}{\text{width}}$ .
- 3 Second bar: width  $25 - 15 = 10$ , so frequency  $= 1.5 \times 10 = 15$ .  
why: Same rule, wider class.
- 4 Add:  $16 + 15 = 31$ .  
why: Total frequency of the two classes.

#### Where it goes wrong

- Adding the heights:  $3.2 + 1.5 = 4.7$  - an offered option.
- Using the same width for both bars; the second is twice as wide even though it is shorter.

**Takeaway** Histogram: frequency = density  $\times$  class width (area of the bar).

### Question 3 STA-03 · Statistics · spec M6.3

The numbers 3, 7,  $x$ , 12, 18 (with  $7 \leq x \leq 12$ ) have mean equal to median. Find  $x$ .

**Answer: C**

The median is  $x$  and the mean is  $\frac{40+x}{5}$ . Setting them equal:  $40 + x = 5x$ , so  $x = 10$ .

#### Full method - how and why

Write expressions for the mean and the median, then equate. The given range for  $x$  tells you where it sits in the order.

- 1 Since  $7 \leq x \leq 12$ , the sorted list is 3, 7,  $x$ , 12, 18, so the median is  $x$ .  
*why:* With five values the median is the third; the constraint is there precisely to fix  $x$ 's position.
- 2 Mean =  $\frac{3+7+x+12+18}{5} = \frac{40+x}{5}$ .  
*why:* Sum the known values first:  $3+7+12+18=40$ .
- 3 Set mean = median:  $\frac{40+x}{5} = x$ .  
*why:* That is the condition stated.
- 4 Solve:  $40 + x = 5x \Rightarrow 4x = 40 \Rightarrow x = 10$ .  
*why:* And 10 does lie in  $[7, 12]$ , so the assumed ordering holds.

#### Where it goes wrong

- Taking the median to be 12 (the middle of the numbers as originally listed without  $x$ ).
- Forgetting to check that the answer is consistent with the assumed position in the order.

**Takeaway** Order first, then write the median; always check the solution respects the ordering you assumed.

### Question 4 STA-04 · Statistics · spec M6.3

The table shows grouped data:  $0 \leq t < 10$ : frequency 4;  $10 \leq t < 20$ : frequency 10;  $20 \leq t < 40$ : frequency 6. Estimate the mean.

**Answer: F**

Use the midpoints 5, 15, 30:  $\frac{4 \cdot 5 + 10 \cdot 15 + 6 \cdot 30}{20} = \frac{20 + 150 + 180}{20} = \frac{350}{20} = 17.5$ .

#### Full method - how and why

Grouped data: you cannot know the individual values, so **estimate** using class midpoints.

- 1 Find midpoints:  $\frac{0+10}{2} = 5$ ,  $\frac{10+20}{2} = 15$ ,  $\frac{20+40}{2} = 30$ .  
*why:* The midpoint is the best single representative of a class.
- 2 Multiply each by its frequency:  $4(5) = 20$ ,  $10(15) = 150$ ,  $6(30) = 180$ .  
*why:* This estimates the total of all the values in each class.
- 3 Total = 350; total frequency =  $4 + 10 + 6 = 20$ .  
*why:* Add both columns.
- 4 Mean =  $\frac{350}{20} = 17.5$ .  
*why:* Divide estimated total by the number of observations.

#### Where it goes wrong

- Using the midpoint 25 for the last class by assuming equal widths - it runs from 20 to 40, so its midpoint is 30.
- Dividing by 3 (the number of classes) instead of 20.

**Takeaway** Grouped mean  $\approx \frac{\sum f x_{\text{mid}}}{\sum f}$  - and read each class width individually.

### Question 5 STA-05 · Statistics · spec M6.4

Which of the following statements about a scatter graph showing strong positive correlation between hours of revision and exam mark is/are valid?

- I Students who revised more tended to score higher.
- II Revising more causes a higher mark.
- III A line of best fit could be used to estimate the mark of a student who revised for a number of hours within the range of the data.

**Answer: F**

Correlation describes a tendency (I) and supports interpolation within the data range (III), but does not by itself establish causation (II).

#### Full method - how and why

A correlation-and-causation statement test. Decide what correlation does and does not licence.

- 1 I: strong positive correlation means higher revision went with higher marks. **True** - that is exactly what correlation describes.

why: 'Tended to' is a statement about the trend, not about every student.

- 2 II: correlation does not establish causation. **False.**

why: A hidden variable (motivation, prior ability) could drive both, so the data alone cannot prove cause.

- 3 III: using a line of best fit **within** the data range is interpolation. **True.**

why: Interpolation is legitimate; extrapolation beyond the data is what is unreliable.

- 4 So I and III only.

why: Match to the list.

#### Where it goes wrong

- Accepting II because the causal story sounds plausible - plausibility is not evidence.
- Rejecting III by confusing interpolation with extrapolation.

**Takeaway** Correlation licenses description and interpolation; it never licenses a causal claim on its own.

### Question 6 STA-06 · Statistics · spec M6.3

A set of  $n$  numbers has mean  $m$ . Each number is doubled and then 3 is added. Which of the following is the new mean, and which the new range (if the original range was  $R$ )?

**Answer: A**

Mean transforms with the data:  $2m + 3$ . The range is unaffected by adding a constant but doubles with the scaling:  $2R$ .

#### Full method - how and why

Coding: see separately how a linear transformation affects a measure of **location** (the mean) and a measure of **spread** (the range).

- 1 Each value  $x$  becomes  $2x + 3$ .  
*why:* Read the transformation carefully: double, then add.
- 2 The mean follows the same transformation: new mean  $= 2m + 3$ .  
*why:* Because  $\overline{2x + 3} = 2\bar{x} + 3$  - the mean is linear.
- 3 The range is a **difference**:  $(2x_{\max} + 3) - (2x_{\min} + 3) = 2(x_{\max} - x_{\min}) = 2R$ .  
*why:* The  $+3$  cancels in the subtraction; only the scaling survives.
- 4 So mean  $2m + 3$ , range  $2R$ .  
*why:* Location shifts and scales; spread only scales.

#### Where it goes wrong

- Giving range  $2R + 3$  - the offered trap; adding a constant to every value shifts the data but does not stretch it.
- Leaving the range unchanged as  $R$ , forgetting the doubling.

**Takeaway** Under  $x \rightarrow ax + b$ : mean  $\rightarrow a\bar{x} + b$ , but spread (range, IQR, SD)  $\rightarrow |a| \times \text{spread}$  - the  $+b$  never affects spread.

### Question 7 STA-08 · Statistics · spec M6.3

A cumulative frequency graph for 80 observations passes through (20, 20), (30, 40) and (45, 60). Estimate the interquartile range.

**Answer: D**

Lower quartile at cumulative frequency 20: value 20. Upper quartile at 60: value 45. IQR = 25.

#### Full method - how and why

On a cumulative frequency graph, quartiles are read by going **in** from the cumulative frequency axis at  $\frac{n}{4}$ ,  $\frac{n}{2}$ ,  $\frac{3n}{4}$ .

- 1 With  $n = 80$ : the lower quartile is at cumulative frequency  $\frac{80}{4} = 20$ , the upper at  $\frac{3 \times 80}{4} = 60$ .  
*why:* Cumulative frequency graphs use  $\frac{n}{4}$  and  $\frac{3n}{4}$  directly (no  $+1$ , unlike a raw list).
- 2 The point (20, 20) gives  $Q_1 = 20$ .  
*why:* Read across at CF = 20 and down to the value axis.
- 3 The point (45, 60) gives  $Q_3 = 45$ .  
*why:* Read across at CF = 60.
- 4 IQR  $= Q_3 - Q_1 = 45 - 20 = 25$ .  
*why:* The interquartile range is a difference of **values**, not of frequencies.

#### Where it goes wrong

- Subtracting the cumulative frequencies,  $60 - 20 = 40$  - an offered option.
- Using the median point (30, 40) as a quartile.

**Takeaway** Go in from the frequency axis at  $\frac{n}{4}$  and  $\frac{3n}{4}$ , read off values, then subtract the **values**.

### Question 8 STA-09 · Statistics · spec M6.3

Class A has 20 students with mean mark 60; class B has 30 students with mean mark 70. What is the mean mark of all 50 students?

**Answer: F**

$$\frac{1200 + 2100}{50} = \frac{3300}{50} = 66.$$

#### Full method - how and why

A combined mean is a **weighted** average: go back to totals, since the classes are different sizes.

1 Class A total =  $20 \times 60 = 1200$ .

why: Total = mean  $\times$  number.

2 Class B total =  $30 \times 70 = 2100$ .

why: Same again for the larger class.

3 Combined mean =  $\frac{1200 + 2100}{50} = \frac{3300}{50} = 66$ .

why: Divide the grand total by the grand number of students.

#### Where it goes wrong

- Averaging the means:  $\frac{60+70}{2} = 65$  - an offered option, and wrong because class B has more students.
- A quick check: the answer must lie **\*\*closer to 70\*\*** than to 60, which 66 does.

**Takeaway** Never average averages; weight by group size - or just work with totals.

### Question 9 STA-10 · Statistics · spec M6.3

For which of the following data sets is the mean greater than the median?

**Answer: D**

A large outlier pulls the mean above the median: mean 6 > median 3. In B, D, E the mean equals the median; in C the mean 9.4 < median 11.

#### Full method - how and why

The mean is pulled by extreme values; the median is not. Look for skew rather than computing everything.

1 Check 1, 2, 3, 4, 20: median = 3, mean =  $\frac{30}{5} = 6$ . Mean > median. **Yes.**

why: The single large outlier 20 drags the mean far above the middle.

2 1, 2, 3, 4, 5 and 2, 4, 6, 8, 10 and 5, 5, 5, 5, 5 are symmetric, so mean = median.

why: For symmetric data the two coincide - no need to compute.

3 1, 10, 11, 12, 13: median = 11, mean =  $\frac{47}{5} = 9.4$ , so mean < median.

why: Here the outlier is small, so it drags the mean **down** - the mirror image.

4 So only 1, 2, 3, 4, 20 works.

why: One high outlier gives positive skew.

#### Where it goes wrong

- Assuming any outlier raises the mean - a low outlier lowers it, as in 1, 10, 11, 12, 13.
- Computing all five means when symmetry settles three of them instantly.

**Takeaway** Mean > median signals a tail to the **right** (a high outlier); mean < median signals a tail to the left.

### Question 10 STA-07 · Statistics · spec M6.3

The median of 7 distinct integers is 10 and the mean is 12. The smallest number is increased by 20. Which of the following must be true?

**Answer: D**

The mean rises by  $\frac{20}{7}$ . The smallest number plus 20 could exceed the median (e.g. if the smallest is 1 and it becomes 21 the middle value changes), or it could stay below it. So the median may or may not change.

#### Full method - how and why

A 'must be true' question. Test each claim by trying to construct a counterexample.

- 1 The mean: the total rises by 20, so the mean rises by  $\frac{20}{7}$  - it **definitely** increases.  
why: Nothing else changed, so the total change is +20 across 7 values.
- 2 The median: try the smallest = 1 becoming 21. That value jumps past the middle, so the values shift along and the median changes.  
why: Moving a value from below the median to above it pushes the middle value up one place.
- 3 Now try the smallest = -15 becoming 5, still below the median 10. The ordering of the middle is untouched, so the median is unchanged.  
why: A value that stays on the same side of the median cannot move it.
- 4 Both are possible, so the median **may or may not** change while the mean definitely rises.  
why: That is precisely the statement offered.

#### Where it goes wrong

- Claiming the median never changes - true only if the increased value stays below the median.
- Claiming both must increase, which would need the smallest to always overtake the middle.

**Takeaway** 'Must be true' means no counterexample exists; hunt for one before agreeing.

**Question 11** X2-28 · Statistics · spec M6.3

A set of five distinct positive integers has median 10 and mean 10. Which of the following statements must be true?

- I The largest value is at least 12.
- II The smallest value is at most 8.
- III The range is at least 4.

**Answer: H**

The two values above the median are distinct integers greater than 10, so at least 11 and 12 (I); similarly the two below are at most 9 and 8 (II); hence the range is at least  $12 - 8 = 4$  (III).

**Full method - how and why**

Extremal reasoning with **distinct integers**: the distinctness forces gaps of at least 1 on each side of the median.

1 Order the five values. The median (third value) is 10.

why: Five values means the middle one is the third.

2 The two above are distinct integers greater than 10, so at least 11 and 12. Hence the largest is  $\geq 12$ . **I true.**

why: Distinct integers above 10 must be 11 or more, and two of them cannot both be 11.

3 Symmetrically, the two below are at most 9 and 8, so the smallest is  $\leq 8$ . **II true.**

why: Same argument downwards.

4 Range  $\geq 12 - 8 = 4$ . **III true.**

why: Combining the two bounds. (The mean being 10 is consistent - e.g. 8, 9, 10, 11, 12 - but is not needed for any of the three claims.)

5 So all three.

why: Match to the list.

**Where it goes wrong**

- Forgetting 'distinct' and allowing two values equal to 11, which would break I.
- Assuming the mean condition must be used; here it is extra information.

**Takeaway** Distinct integers force minimum gaps:  $k$  values above a median  $m$  reach at least  $m + k$ .